

$$\mathbf{Y}_{t,k}^X = \begin{bmatrix} \mathbf{Y}_{t,k}^Z \\ \mathbf{Y}_t^A \end{bmatrix}, \quad \mathbf{W}_{t,k}^X = \begin{bmatrix} \mathbf{W}_{t,k}^Z & \mathbf{0} \\ \mathbf{0} & \mathbf{W}_t^A \end{bmatrix},$$

$$\mathbf{P}^X = \text{blkdiag}(P_Z \mathbf{I}_{D_Z}, P_A \mathbf{I}_{D_A})$$